

- All workings must be clearly shown.

KENYATTA UNIVERSITY
ECV 211: ELECTROMECHANICAL ENGINEERING

DATE: 16TH DECEMBER 2021

TIME: 1 HOURS

-
- 1) Differentiate between active and passive solar water heating systems and give applications where each is best suited. **[6 marks]**
 - 2)
 - a) Give reasons why thermoplastic PVC is the most commonly used type of insulation today
 - b) What is the greatest shortcoming of PVC and how has this been handled in modern applications **[5 marks]**
 - 3) Fire protection is a key feature in building services. Give THREE main systems that can be installed to alert occupants in case of fire at their premises **[6 marks]**
 - 4) With the aid of a well labelled diagram explain how a two way light switch operates **[5 marks]**
 - 5) Explain the functions performed by each of the following components used in domestic water heating systems. Also explain where they are normally placed.
 - a) Air Vent
 - b) Isolation Valves
 - c) Temperature-Pressure Relief Valve
 - d) Check Valves **[8 marks]**

KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN CIVIL
ENGINEERING

ECV 211: ELECTROMECHANICAL ENGINEERING

1ST SEMESTER 2021/2022

TIME: 2Hours

INSTRUCTIONS:

- This paper consists **FIVE** questions. Answer **Question ONE** and any other **TWO Questions**
- All workings must be clearly shown.

QUESTION ONE (30 MARKS)

a) Using sketches explain the difference between a short circuit and an open circuit.

[4 marks]

b)

- i) State and briefly explain the Kirchoff's laws as applied in circuit electrical circuit analysis.
- ii) Using the stated laws, determine the currents flowing in each branch of the network shown in Figure Q 1(b).

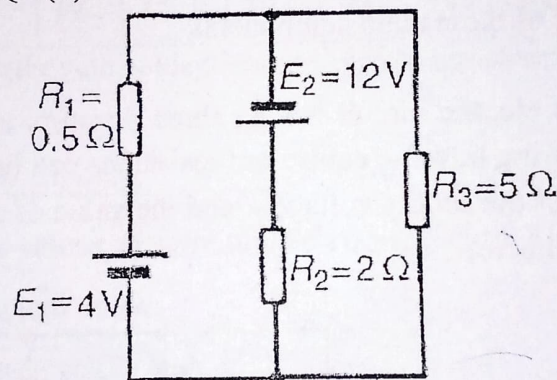


Figure Q1 (b)

[14 marks]

c) Capacitance's of $1\ \mu\text{F}$, $3\ \mu\text{F}$, $5\ \mu\text{F}$ and $6\ \mu\text{F}$ are connected in parallel to a direct voltage supply of 100V . Determine:

- i) The equivalent circuit capacitance,
- ii) the total charge and
- iii) the charge on each capacitor.

[6 marks]

d) Explain reasons why domestic solar water heating is highly encouraged as part of electricity conservation measure in building services

[6 marks]

QUESTION TWO (20 MARKS)

- Briefly discuss the main requirements for the conversion of mechanical energy to electrical energy . [3 marks]
- Briefly discuss any FOUR types of insulations which can be used in electrical cables. [8 marks]
- A generator supplies power to an electric heater, an inductive element, and a capacitor as in Figure Q3(a).

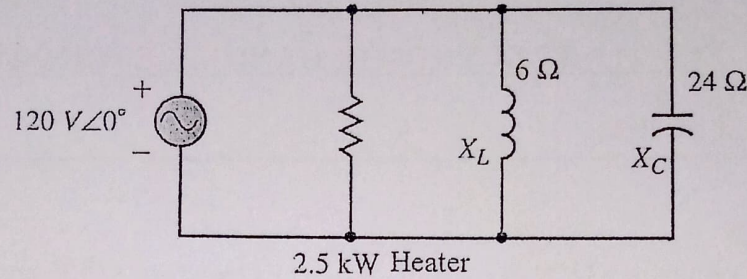


Figure Q3(a)

- Find P and Q for each load.
- Find total active and reactive power supplied by the generator.
- Draw the power triangle for the combined loads and determine total apparent power.
- Find the current supplied by the generator.

[9 marks]

QUESTION THREE (20 MARKS)

- With the aid of schematics show the typical components of a direct flat plate collector system used in domestic water heating.
 - Explain the use of these main components
- Considering an electric circuit having three resistors arranged in parallel, derive an expression showing how the equivalent resistance can be obtained.
 - Find the value of the supply voltage V and the value of current I for the circuit shown in Figure Q3 (b) below,

[13 marks]

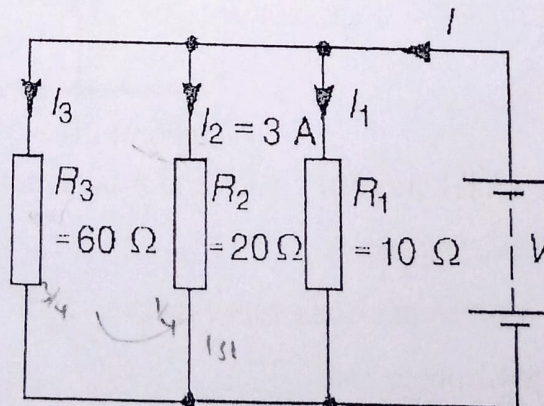


Figure Q3 (b)

[7 marks]

Thermocouple PVC
butyl rubber
Silicone rubber
Glass
paper

Handwritten calculations for Figure Q3 (b):
 $15 \times 2 = 30$
 $30 = 30$
 $30 \times 10 = 300$
 $10 \times 30 = 300$
 $300 = 300$
 $15 + 10 = 25$
 $10/25 = 2/5$

QUESTION FOUR (20 MARKS)

a) A $30\mu\text{F}$ capacitor is connected in parallel with an 80 ohms resistor across a 240V, 50 Hz supply. Calculate:

- i) the current in each branch,
- ii) the supply current,
- iii) the circuit phase angle,
- iv) the circuit impedance,
- v) the true or active power dissipated
- vi) the apparent power

[12 marks]

b)

- (i) What is a measurement instrument?
- (ii) Differentiate between basic and derived measurement units

[4 marks]

c) An e.m.f. of 1.5 kV is induced in a coil when a current of 4A collapses uniformly to zero in 8 ms. Determine the inductance of the coil.

[4 marks]

QUESTION FIVE (20 MARKS)

a) The following table gives the corresponding values of current and time for a half cycle of alternating current.

Time (ms)	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
Current (A)	0	7	14	23	40	56	68	76	60	5	0

Assuming the negative half cycle is identical in shape to the positive half cycle, plot the waveform and find:

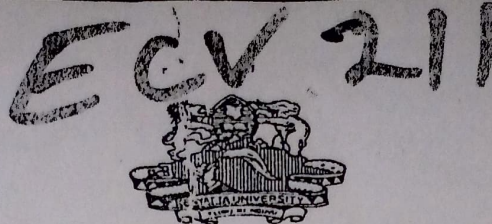
- i) the frequency of the supply,
- ii) the instantaneous values of current after 1.25 ms and 3.8 ms,
- iii) the peak or maximum value,
- iv) the mean or average value, and
- v) the r.m.s. value of the waveform.

(13 mark)

b) Define impedance and hence give the formulas used in obtaining resistance, inductance and capacitance impedances.

[7 marks]

**** END OF EXAM ****



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2013/2014
SECOND SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE (CIVIL ENGINEERING)
ECV 211: ELECTROMECHANICAL ENGINEERING

DATE: Monday, 14th April 2014

TIME: 11.00 a.m. – 1.00 p.m.

INSTRUCTIONS:

1. This paper consists of FIVE questions.
2. Attempt Question ONE (compulsory), and any other TWO questions.
3. Question ONE contains 30 (THIRTY) marks. All other questions contain 20 (TWENTY) marks.
4. The charge of an electron is given as $1.6 \times 10^{-19} \text{C}$.

SECTION A (COMPULSORY)

Question ONE (30 Marks)

- (a) Define the following terms:
- (i) Storm water
 - (ii) Fire stop
 - (iii) Coincidence factor
 - (iv) Shunt
 - (v) Conductance
- (5 marks)
- (b) Explain the three main tests conducted on a completed electrical installation before commissioning.
- (6 marks)
- (c) List three requirements for a fire refuge area in a building.
- (3 marks)
- (d) Outline two advantages and two disadvantages of moving iron type meters.
- (4 marks)
- (e) Four resistors of equal value are connected in parallel. The effective total resistance of the combination is 14Ω . Calculate the value of each resistor.
- (3 marks)
- (f) Describe the operation of a direct expansion coil.
- (3 marks)
- (g) State two advantages of incinerator systems
- (2 marks)
- (h) Derive the expression for the voltage divider formula.
- (4 marks)

SECTION B: Answer any TWO Questions from this section
Question TWO (20 Marks)

- (a) A commercial building has a floor area of 485 m^2 . The cable length for the sockets and lighting circuits has been calculated to be 420 m . Calculate the minimum number of ring circuits that need to be set up within the building. (6 marks)
- (b) With the aid of a labelled diagram, describe the variable air volume system used in air conditioning. (8 marks)
- (c) Derive a simplified expression for the equivalent resistance of three unequal resistors R_1 , R_2 and R_3 connected in parallel. (6 marks)

Question THREE (20 marks)

- (a) An assembly hall has a capacity of 1800 people. If each door out of the hall has a width of 1875 mm, calculate the number of exit doors required in order to ensure safe evacuation in case of a fire. (10 marks)
- (b) The circuit in Figure Q3 consists of six resistors, each of value 150Ω . Calculate the current flowing through the resistor labelled R_x if the voltage between A and B is 24 V . (10 marks)

Question FOUR (20 marks)

- (a) A hospital is to be built with elevators with a carrying capacity of 45 passengers per elevator car. The peak traffic has been estimated at 350 passengers per minute during the busiest operating times. The elevator round trip time has been estimated at 3 minutes. Calculate the minimum number of elevators required to serve the hospital. (8 marks)
- (b) A circuit consisting of a 10Ω resistor and a 15 mH inductor in series is connected to a 12 V ac supply operating at 150 Hz . Calculate the number of electrons flowing through a point in the circuit in 1 minute. (8 marks)
- (c) List four considerations used in determining pump ratings. (4 marks)

Question FIVE (20 marks)

- (a) With the aid of a well labelled diagram, describe the operation of a macerator system. (10 marks)
- (b) An ohmmeter is to be designed from a moving coil meter with a full scale deviation of 5 mA and an internal resistance of 200Ω . Given that the half-scale deviation is to be $10 \text{ k}\Omega$ and the three-quarter scale deviation is to be $2 \text{ k}\Omega$, calculate the voltage and series resistance required. (10 marks)



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2013/2014
SECOND SEMESTER EXAMINATION FOR THE DEGREE OF
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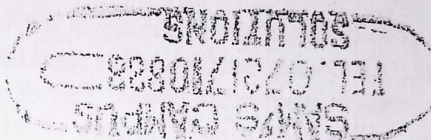
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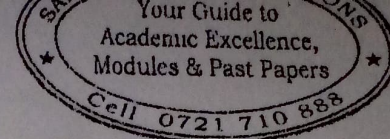
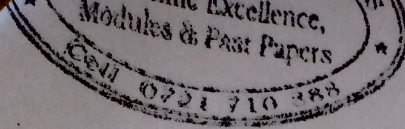
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KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (CIVIL ENGINEERING)

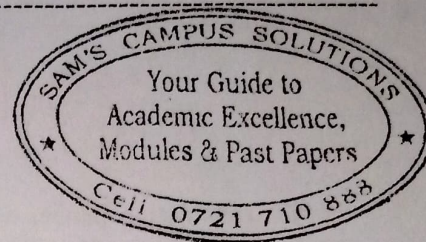
ECV 211: ELECTROMECHANICAL ENGINEERING

DATE: WEDNESDAY 22ND AUGUST 2018

TIME: 8.00 A.M. - 10.00 A.M.

INSTRUCTIONS

1. This paper consists of FIVE questions.
2. Attempt QUESTION ONE and any other TWO questions.
3. All workings must be clearly shown.



QUESTION ONE (30 Marks)

- a) Electrical equipment in an office takes a current of 13A from a 240V supply. Estimate the cost per week of electricity if the equipment is used for 30 hours each week and 1 kWh of energy costs Ksh 13.56 [4 marks]
- b) Briefly explain FOUR benefits of domestic solar water heating? [6 marks]
- c)
- (i) Explain how a capacitor is discharged
 - (ii) Using a circuit diagram derive an expression showing how the effective capacitor resistance of 3 capacitors connected in series can be obtained. [6 marks]
- d) Find the amplitude, phase, period, and frequency of the sinusoid $v(t) = 12 \cos(50t + 10)$ [4 marks]
- e) Define the term Inductance and hence differentiate between self-inductance and mutual inductance [3 marks]
- f) Explain why a ring circuit is preferred to a radial circuit in electrical installations [2 marks]
- g) Highlight some of the reasons for having a fire protection in buildings [5 marks]

capacitor resistance
not resistors

QUESTION TWO (20 Marks)

- a) Find the currents and voltages in the circuit shown in Figure Q2 (a)

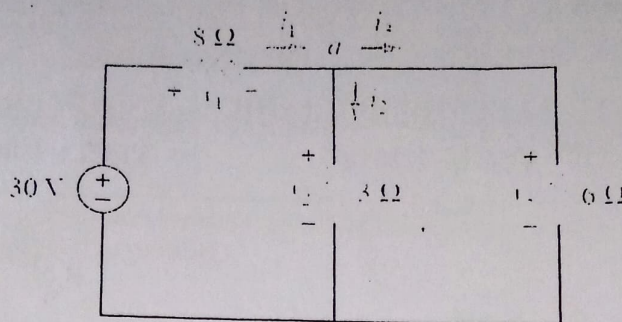


Figure Q2(a)

[11 marks]

- b) Discuss the properties of these types of insulations used in electrical cables ✓
- (i) Thermoplastic PVC
 - (ii) Thermosetting insulation
 - (iii) Butyl rubber
 - (iv) Silicone rubber
 - (v) Glass
 - (vi) Paper

[9 marks]

QUESTION THREE (20 Marks)

- a) Explain the meaning of the following terms with respect to sinusoidal AC quantities. Give examples where necessary
- i. Instantaneous values - value of AC at any instant time
 - ii. Peak Values - largest value measured in half cycle
 - iii. Average Value - Average value measured in half cycle
 - iv. Effective Value - Value of AC to produce same heating as an equivalent DC
 - v. Form factor - Ratio of rms to avg value
 - vi. Peak factor - Ratio of max to rms

[9 marks]

- b) Explain why all electrical installations require proper earthing.

[4 marks]

- c) In the circuit given below, $C_1=60\mu\text{F}$, $C_2=20\mu\text{F}$, $C_3=9\mu\text{F}$ and $C_4=12\mu\text{F}$. If the potential difference between points a and b ($V_{ab}=120\text{V}$) find the charge of the second capacitor.

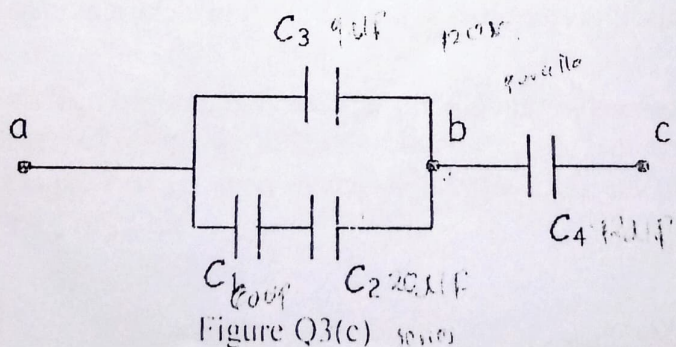


Figure Q3(c)

[7 marks]

Resistance
series = $R_1 + R_2$
Capacitance
series = $\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$
 $V = \left(\frac{P_1}{R_1 + R_2} \right)$

KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
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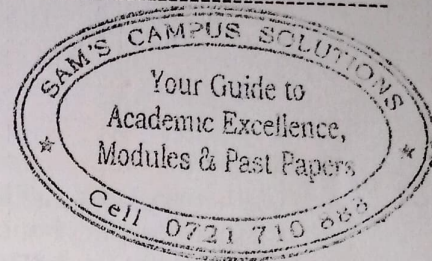
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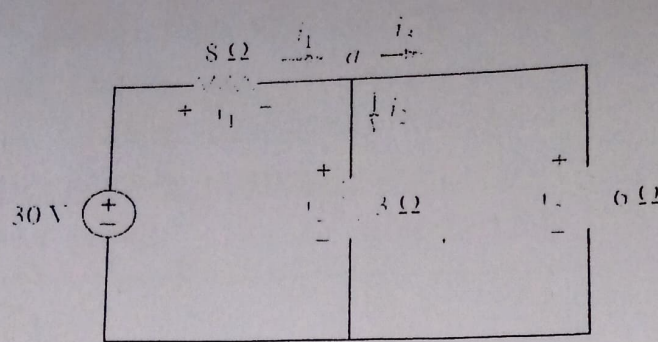


Figure Q 2(a)

[11 marks]

b) Discuss the properties of these types of insulations used in electrical cables

- (i) Thermoplastic PVC $\rightarrow 70^\circ$
- (ii) Thermosetting insulation $\times$ XLPE
- (iii) Butyl rubber $\rightarrow 85^\circ$
- (iv) Silicone rubber $\rightarrow 60 - 150^\circ$
- (v) Glass $\rightarrow 200^\circ$
- (vi) Paper $\rightarrow$ underground low/medium voltage transmission

[9 marks]

QUESTION THREE (20 Marks)

a) Explain the meaning of the following terms with respect to sinusoidal AC quantities. Give examples where necessary

- i. Instantaneous values
- ii. Peak Values
- iii. Average Value
- iv. Effective Value
- v. Form factor
- vi. Peak factor.

[9 marks]

b) Explain why all electrical installations require proper earthing.

[4 marks]

c) In the circuit given below, $C_1=60\mu\text{F}$, $C_2=20\mu\text{F}$, $C_3=9\mu\text{F}$ and $C_4=12\mu\text{F}$. If the potential difference between points a and b ($V_{ab}=120\text{V}$) find the charge of the second capacitor.

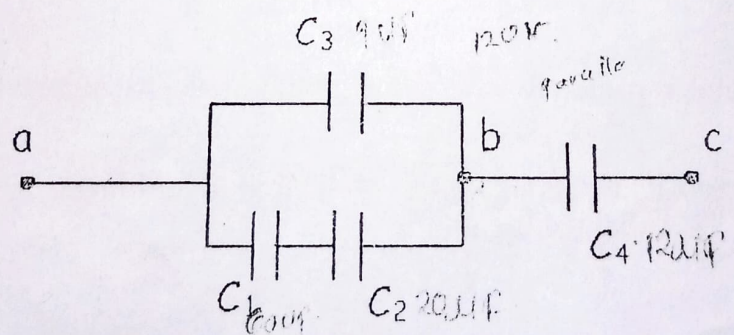


Figure Q3(c)

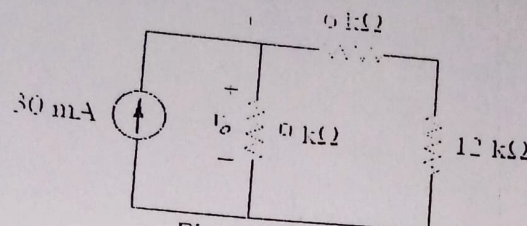
[7 marks]

211
22nd August 2018

QUESTION FOUR (20 Marks)

- a) For the circuit shown in Figure Q4 (a), determine:
- (i) the voltage v_o ,
 - (ii) the power supplied by the current source,
 - (iii) the power absorbed by each resistor.

[9 marks]



- b) Using a schematic diagram show how two way switching is achieved [7 marks]
- c) What are the voltages that are used in generation, transmission, primary distribution and domestic distribution with respect to the Kenyan power system [4 marks]

QUESTION FIVE (20 Marks)

- a)
- (i) What is a measurement instrument?
 - (ii) Differentiate between basic and derived measurement units
- [4 marks]
- b) With the aid of a well labelled schematic diagram briefly explain the parts of a domestic electricity supply system from entry to sub-circuits.
- [6 marks]
- c)
- (i) With the aid of schematics show the typical components of a direct flat plate collector system used in domestic water heating.
 - (ii) Explain the use of these main components

[10 marks]

22nd August 2018

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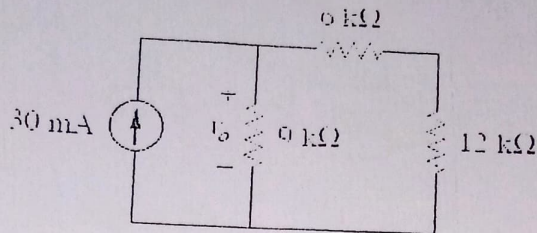


Figure Q4 (a)

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 - Explain the use of these main components

[6 marks]

[10 marks]

14 April 2014

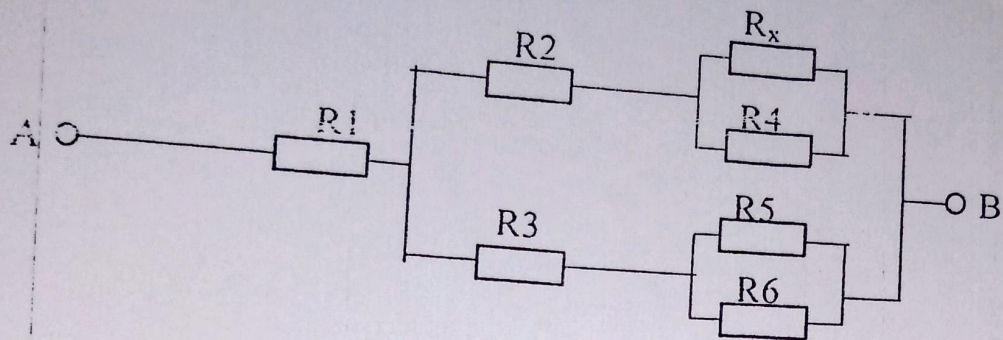


Figure Q3

I_a

$I_T =$

$Y_0 = \frac{I_T}{V_0}$